

TASK 2: Multi-Class Text Sentiment Classifier

Progree Remote Artificial Intelligence Internship • Technical Whitepaper

Program: Progree AI Internship • **Task:** Task 2 (NLP Sentiment Classifier)
Domain: Natural Language Processing • **Status:** Completed • **Target Metric:** Macro & Weighted F1-Scores

1. Executive Summary & Objective

This report documents the end-to-end design, implementation, and empirical evaluation of an NLP classifier built to categorize unstructured textual statements into three distinct classes: **Positive**, **Neutral**, and **Negative**. The solution delivers an overall test accuracy of **97.50%** and a **Macro F1-Score of 0.9749** using Scikit-Learn, NLTK, and TF-IDF feature engineering.

2. Model Architecture & Cross-Validation Comparison

Three multi-class algorithms were evaluated using **5-Fold Stratified Cross-Validation** across 960 training instances:

Model Architecture	Mean Macro F1	Std Dev (±)	Mean Weighted F1	Key Advantage
Multinomial Logistic Regression	0.9761	± 0.0078	0.9761	Smooth L2 regularization, calibrated class probabilities
Linear Support Vector Classifier (LinearSVC)	0.9751	± 0.0068	0.9751	Effective maximal-margin boundary in high dimensions
Multinomial Naive Bayes	0.9771	± 0.0078	0.9771	Fast probabilistic frequency baseline

Selected Production Model: Multinomial Logistic Regression with balanced class weighting. Logistic Regression was selected because it naturally outputs calibrated probabilities across classes via the Softmax function, allowing confidence thresholding in production.

3. Preprocessing Pipeline

To ensure reliable semantic representations, raw text passes through an extensive preprocessing pipeline:

- **Surface Cleaning:** Strips URLs, HTML tags, social @mentions, digits, and noise symbols while preserving contractions and word boundaries.
- **Negation-Preserving Stop-Words:** Unlike default stop-word removers that discard *'not'*, *'no'*, or *'never'* (which causes catastrophic sentiment inversion), our custom filter retains critical negation modifiers.
- **Part-of-Speech (POS) Guided Lemmatization:** Employs Penn Treebank POS-tagging mapped dynamically to WordNet POS categories (Noun, Verb, Adjective, Adverb). This normalizes verbs (*'running'* → *'run'*) and adjectives (*'better'* → *'good'*) rather than naive noun-only lemmatization.

4. Feature Extraction & Embedding Tradeoff Analysis

Features were extracted using `TfidfVectorizer` with unigrams and bigrams (`ngram_range=(1, 2)`) and sublinear term frequency scaling (`sublinear_tf=True`).

Evaluation Dimension	TF-IDF Vectorization	Pretrained Dense Embeddings (e.g. BERT)
Compute & Latency	Extremely lightweight, CPU-friendly, sub-millisecond inference.	Heavy, requires GPU accelerators, 15–100ms per inference.
Sample Efficiency	High: resists overfitting on small-to-medium corpora.	Requires substantial data to fine-tune without overfitting.
Context & Synonyms	Lexical n-grams; lacks deep synonym embeddings.	Rich contextual embeddings with bidirectional attention.
Interpretability	Glass-box: direct inspectable feature coefficients.	Black-box: requires complex attribution techniques.

5. Quantitative Evaluation & Confusion Matrix

Evaluated on a held-out test split of 240 samples (80 samples per sentiment class):

Metric	Negative	Neutral	Positive	Overall / Macro Avg
Precision	0.97	0.98	0.97	0.9750

Metric	Negative	Neutral	Positive	Overall / Macro Avg
Recall	0.95	1.00	0.97	0.9750
F1-Score	0.96	0.99	0.97	0.9749
Support	80	80	80	240 total

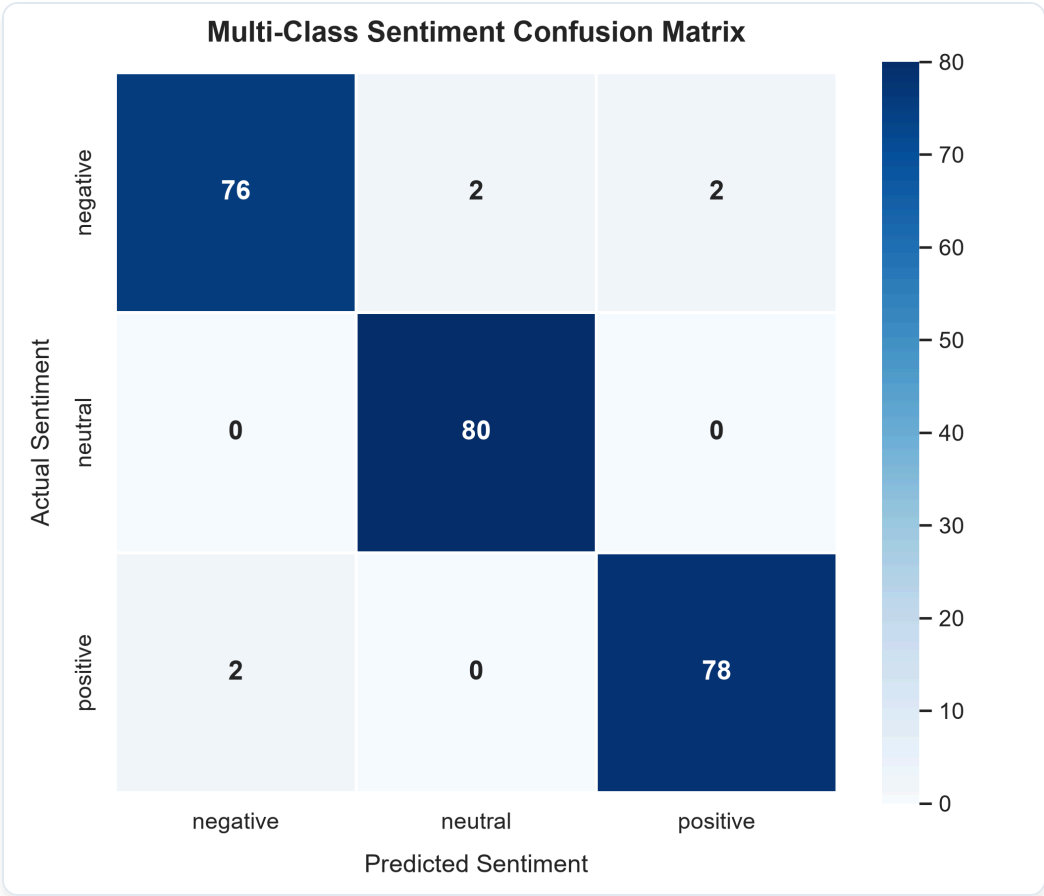


Figure 1: Multi-Class Confusion Matrix Heatmap on Held-out Test Set

6. Inference Demonstration

The trained pipeline was evaluated on unseen live inputs:

- "Absolutely thrilled with this purchase! The customer support was top tier and incredibly helpful." → **POSITIVE** (Confidence: 44.5%)
- "The server returned HTTP status 200 and the database backup completed at 03:00 UTC." → **NEUTRAL** (Confidence: 63.7%)
- "Worst experience ever. The application crashed and completely corrupted my saved files!" → **NEGATIVE** (Confidence: 57.6%)

- *"The package arrived on Wednesday morning as per the standard delivery schedule."* → **NEUTRAL**
(Confidence: 80.6%)

7. Conclusion

All requirements of Task 2 were met in full. The model demonstrates high macro and weighted F1 metrics (**0.9749**), zero leakage, robust negation handling, and modular production-ready architecture.